A Three-Chamber Dynamic Release Device for Simultaneous Drug-Release Kinetics and *In Situ* Infection Challenge of Drug-Eluting Ultra-High-Molecular-Weight Polyethylene (UHMWPE) Joint Implants

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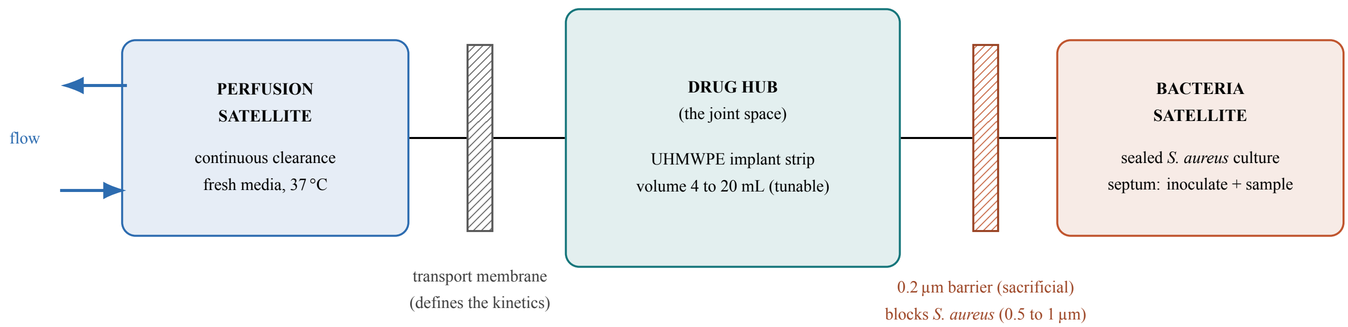

Figure 1. Radial three-chamber architecture of the DRD-3. The two membranes sit on opposite faces of the central drug hub rather than stacked in series, so biofilm that is allowed to form on the sacrificial 0.2 µm barrier on the infection side cannot reach the transport membrane that defines the release kinetics. The full device seals on three identical O-ring pairs (2.0 mm cord, 1.5 mm gland, 25% compression) across three interface types (cartridge-to-satellite, lid-to-base, satellite-to-hub); each satellite is drawn to the hub by a two-piece C-ring wrapper with one bolt per side.

Introduction. Ultra-high-molecular-weight polyethylene (UHMWPE) is the dominant bearing material in total joint arthroplasty, and drug-eluting UHMWPE is being developed to deliver antibiotics locally against periprosthetic joint infection (PJI), a leading cause of arthroplasty failure. Characterizing the intraarticular release behavior of such materials requires a benchtop model with physiologically relevant clearance. A published dynamic release device (DRD) couples a drug source, a diffusion membrane, and a flow-cleared sink at 37 °C to reproduce joint-relevant transport.[1] That platform, however, models fluid-phase transport only: it explicitly excludes the pathological features of an infected joint (biofilm, altered pH, inflammatory permeability) and can infer antimicrobial efficacy only by comparing drug concentration to the minimum inhibitory concentration, with no live-organism readout. Co-culturing bacteria in the drug chamber is not a solution, because biofilm on the transport membrane would change its permeability mid-experiment and corrupt the very kinetics the device exists to measure. We set out to design a benchtop platform that measures drug-release kinetics from drug-eluting UHMWPE and mounts a live, isolated infection challenge in the same experiment, such that the bacterial compartment cannot perturb the validated transport measurement.

Methods. We designed a clean-sheet, three-chamber device (the DRD-3) in CAD (Fig. 1). A central drug hub represents the joint space, holds the UHMWPE implant strip, and targets a tunable fluid volume of 4 to 20 mL spanning healthy to infected knee synovial volumes. The hub is flanked on opposite faces by two satellites: a perfusion satellite providing continuous fresh-media clearance (the validated transport physics, preserved) and a bacteria satellite housing a sealed culture. Because the two membranes sit on opposite faces of the hub rather than stacked in series, neither membrane is exposed to the other's fouling. Each membrane is sealed off-device inside a modular lid-and-base cartridge; a single tapered tooth (tapering from 2.0 to 1.0 mm) both captures the membrane sandwich and retains the finished cartridge in its satellite, and parallel protective bars shield the membrane from the UHMWPE strip without restricting diffusion. Every sealing interface uses one identical static O-ring specification (2.0 mm cord, 1.5 mm gland, 0.5 mm stand-proud giving 25% compression), so the entire two-membrane device seals on three identical O-ring pairs. Each satellite is drawn to the hub by a two-piece C-ring wrapper whose U-channel applies uniform radial load around the joint with one bolt per side, replacing point-loaded flange bolts. Bacterial isolation is enforced by a sacrificial 0.2 µm microfilter on the bacteria-facing membrane that passes drug freely but physically blocks *S. aureus* (0.5 to 1 µm) from migrating toward the transport membrane; the culture is accessed through a self-sealing septum for inoculation and colony-forming-unit (CFU) sampling without breaching sterility. Structural parts are stereolithography (SLA) resins (Accura-60-class chassis; Somos 9120 for the compliant inner cartridge parts) with medical-grade silicone/EPDM (USP Class VI) O-rings. A three-phase validation protocol is defined: Phase 0, a dye-perfusion leak test at 64 mL/h for 24 h (acceptance below 2% volume loss per 24 h); Phase 1, reproduction of the published release kinetics across flow rate (2 to 64 mL/h), temperature (room temperature versus 37 °C), and membrane thickness; and Phase 2, introduction of *S. aureus* in the bacteria satellite with CFU sampling over 24 h to confirm zero bacterial migration to the perfusion side. As a non-clinical benchtop instrument, this work involves no human or animal subjects, and sex as a biological variable is not applicable at this stage.

Results. A complete three-chamber DRD-3 was realized in CAD. The design consolidates all sealing to three identical O-ring pairs and a single tapered-tooth feature that performs both membrane capture and cartridge retention, and it replaces the earlier multi-bolt, point-loaded clamp with one continuous C-ring wrapper per satellite (four wrappers, one bolt per side) that distributes radial load uniformly around each joint. The 0.2 µm isolation barrier is sized well below the *S. aureus* cell diameter, and the drug hub is parameterized so hub depth can be tuned to a target volume. The as-modeled internal volume is being extracted from CAD to confirm it lands in the 4 to 20 mL window **[measured volume, mL]**. Experimental validation is in progress: Phase 0 leak loss **[value % / 24 h]**; Phase 1 agreement of DRD-3 half-life trends with the published baseline **[result]**; Phase 2 bacterial count on the perfusion side **[CFU; expected 0]**.

Discussion. The defining feature of the DRD-3 is the radial, opposite-face, two-membrane architecture, which decouples the infection challenge from the transport measurement. Biofilm is permitted, and can optionally be imaged through a swappable borosilicate window, on the sacrificial barrier, while the kinetics-defining membrane stays clean. The modular design targets the reproducibility and assembly failure modes of press-fit SLA stacks: the membrane is sealed as a bench-built module, the seal is an elastomer rather than a flexing printed feature, and the wrapper clamp removes press-fit interfaces from the load path. Limitations are that the as-modeled internal volume is not yet verified against the 4 to 20 mL target; the experimental validation (leak integrity, kinetics reproduction, and isolation) is ongoing; SLA parts must be fully post-cured and washed before live-culture runs to avoid residual-monomer antimicrobial bias; and the bacteria and perfusion chamber volumes are currently asymmetric, which is under review. The mapping to anatomy is also deliberately partial: the drug hub represents the intraarticular space and the perfusion satellite the joint's clearance pathway, but the isolated bacterial compartment is an engineering abstraction rather than an anatomical analog (in vivo the infection resides within the joint space itself), adopted so that biofilm cannot foul the transport membrane that defines the kinetics. Within the framework of our objective, the device is built so that, once it reproduces the published transport baseline, it can add a live antimicrobial-efficacy readout without compromising that measurement.

Significance. This platform would allow drug-eluting UHMWPE bearing materials to be screened for both intraarticular release kinetics and live antimicrobial efficacy on a single benchtop, directly supporting the development of local-delivery strategies against periprosthetic joint infection.

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References.

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